

14 Given the simultaneous equations

$$2x - y = 11,$$

$$5y - 2px + 7 = 0.$$

- (a) Show that $x = \frac{24}{5-p}$.

Answer

[2]

- (b) State the value of p if there is no solution for the pair of simultaneous equations.

Answer $p = \dots\dots\dots$ [1]

- (c) Find the value of y if $p = 1$.

Answer $y = \dots\dots\dots$ [1]

- 15 (a)** Solve the inequality $\frac{1}{3}(4x-5) \leq 1 - \frac{1}{5}x$, and illustrate your solution on the number line given in the answer space.

Answer  [3]

- (b)** Hence, list all possible whole numbers of the solution.

Answer [1]

- 16** An interior angle of a regular n -sided polygon is larger than its exterior angle by 50%. Find n .

Answer $n =$ [2]

17 $\xi = \{x : x \text{ is a positive integer less than } 14\}$

$A = \{x : x \text{ is a composite number}\}$

$B = \left\{x : \frac{x}{3} \text{ is an integer}\right\}$

$C = \{x : x \text{ is a multiple of } 6\}$

(a) List the elements in

(i) A' ,

Answer [1]

(ii) $A' \cap C$.

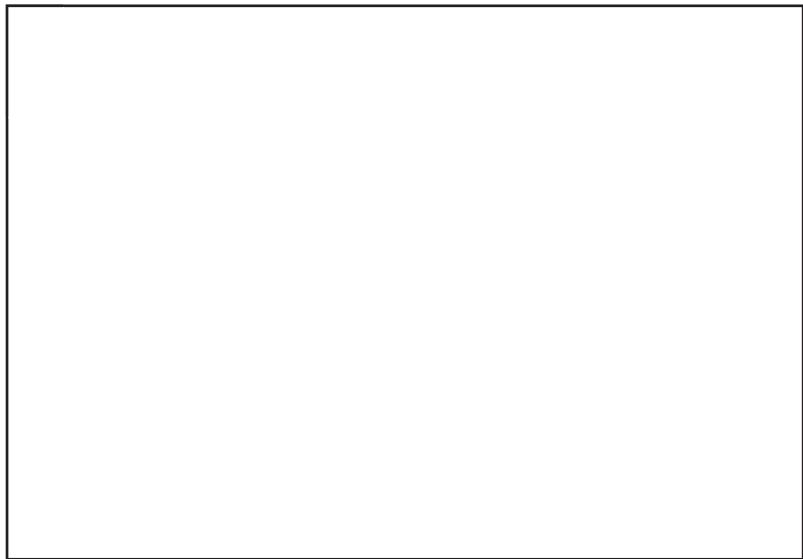
Answer [1]

(b) Draw a Venn diagram to represent the sets A and B .

Label all the elements in $A \cap B$.

Answer

ξ



[2]

(c) List all the proper subsets of set C .

Answer [1]